Overview

- This note summarizes the PD designs used in Lab 3 for the hummingbird simulation and records the specific formulas used to compute gains from time-domain specs. All math is LaTeX to render correctly when exported to PDF (tested with pandoc and typical VS Code PDF exporters that support MathJax/KaTeX).

Models

- Longitudinal (about $\theta_e = 0$):

$$\begin{aligned} \tilde{\theta}''(t) &= b_\theta\, \tilde{F}(t), \\ b_\theta &= \frac{\ell_T}{m_1 \ell_1^2 + m_2 \ell_2^2 + J_{1y} + J_{2y}}, \\ F_e &= \frac{(m_1 \ell_1 + m_2 \ell_2)\, g}{\ell_T}. \end{aligned}$$

- Lateral (successive loop closure, inner roll faster than outer yaw):

$$\begin{aligned} \phi''(t) &= \frac{1}{J_{1x}}\, \tau(t), \\ \psi''(t) &\approx b_\psi\, \phi(t), \quad b_\psi \approx \frac{\ell_T F_e}{J_{zz,\mathrm{eq}}}, \\ J_{zz,\mathrm{eq}} &= m_1 \ell_1^2 + m_2 \ell_2^2 + J_{2z} + J_{1z} + m_3(\ell_{3x}^2 + \ell_{3y}^2) + J_{3z}. \end{aligned}$$

Dirty Derivative (First-Order Filtered Difference)

For any signal x:

$$\begin{aligned} \hat{\dot{x}}[k] &= \alpha\, \hat{\dot{x}}[k-1] + (1-\alpha)\, \frac{x[k] - x[k-1]}{T_s}, \\ \alpha &= \frac{\tau_d}{\tau_d + T_s}. \end{aligned}$$

Gain Mapping from Specs

With desired ζ and rise time t_r:

$$\omega_n \approx \frac{2.2}{t_r}.$$

- Longitudinal (pitch):

$$k_{P_\theta} = \frac{\omega_{n\theta}^2}{b_\theta}, \quad k_{D_\theta} = \frac{2\zeta_\theta\, \omega_{n\theta}}{b_\theta}.$$

- Inner roll:

$$k_{P_\phi} = \omega_{n\phi}^2 J_{1x}, \quad k_{D_\phi} = 2\zeta_\phi\, \omega_{n\phi} J_{1x}.$$

- Outer yaw:

$$k_{P_\psi} = \frac{\omega_{n\psi}^2}{b_\psi}, \quad k_{D_\psi} = \frac{2\zeta_\psi\, \omega_{n\psi}}{b_\psi}.$$

Bandwidth Separation

Choose inner-loop roll at least one order faster than yaw:

$$\omega_{n\phi} = M\,\omega_{n\psi}, \quad M \in [10, 20].$$

Control Laws (PD Only)

- Longitudinal PD (force):

$$\begin{aligned} e_\theta &= \theta_d - \theta, \\ \hat{\dot{e}}_\theta &= -\hat{\dot{\theta}}, \\ \tilde{F} &= k_{P_\theta} e_\theta + k_{D_\theta}\,\hat{\dot{e}}_\theta, \\ F &= F_e + \tilde{F}. \end{aligned}$$

- Lateral PD (successive loops):
  Outer yaw (commands roll reference):

$$e_\psi = \psi_d - \psi, \quad \hat{\dot{e}}_\psi = -\hat{\dot{\psi}}, \quad \phi_d = k_{P_\psi} e_\psi + k_{D_\psi}\,\hat{\dot{e}}_\psi.$$

Inner roll (commands torque):

$$e_\phi = \phi_d - \phi, \quad \hat{\dot{e}}_\phi = -\hat{\dot{\phi}}, \quad \tau = k_{P_\phi} e_\phi + k_{D_\phi}\,\hat{\dot{e}}_\phi.$$

Simulation Notes

- H.7: Step θ_d by $\pm 5^\circ$ and verify well-damped tracking. The logged force F should equal F_e plus PD action.
- H.8: Step ψ_d (e.g., 30°). The roll response should be visibly faster; the rise-time ratio should be near M.

How to Run

From the repository root (recommended to ensure imports resolve):

```
python -m _hummingbird_sim.h7_hummingbirdSim
python -m _hummingbird_sim.h8_hummingbirdSim
```

Or from the directory itself (matches current import style):

```
cd _hummingbird_sim
python h7_hummingbirdSim.py
python h8_hummingbirdSim.py
```

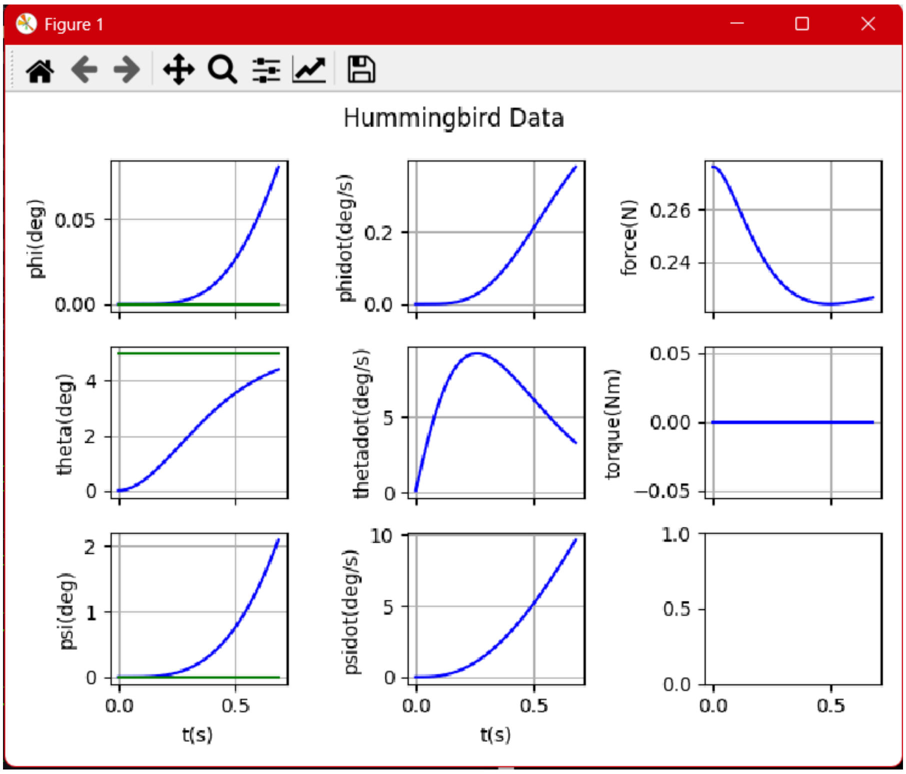
Python version used during development: `3.13`. If matplotlib/pyqt are missing, install from the repo root:

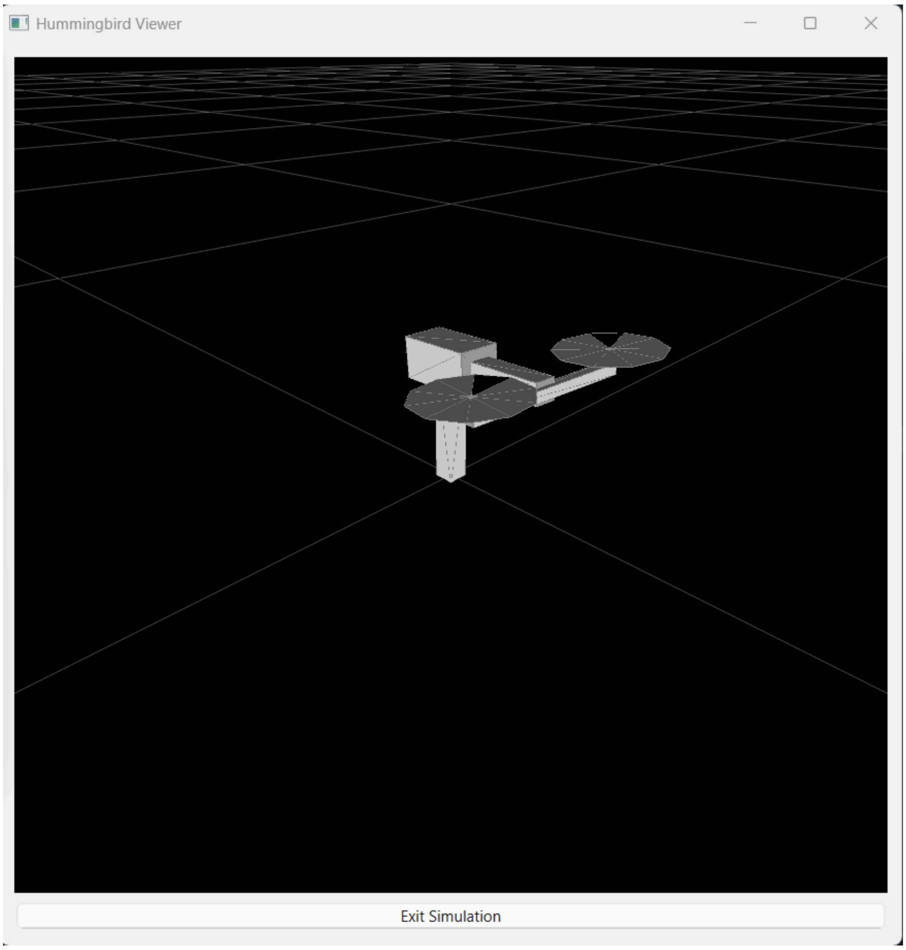
```
python -m pip install --user -r requirements.txt
```

Results and Figures (placeholders)

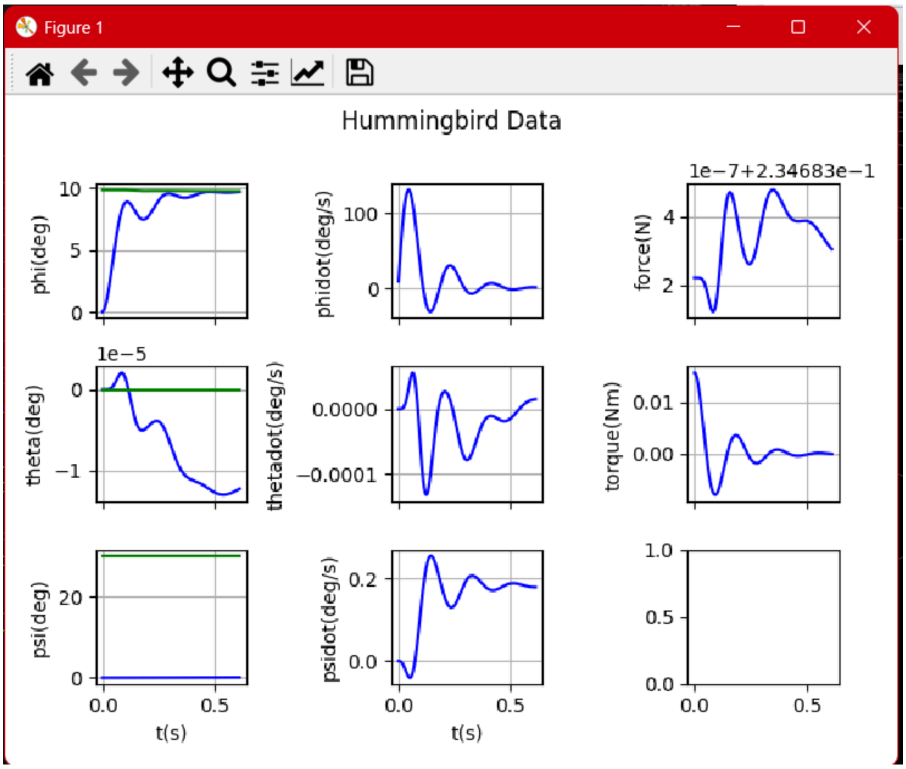
GUI Simulation Output from H.7 & H.8
Images are expected in `../figs/` relative to this Markdown file (as you currently have in the repo). The YAML header sets a LaTeX `\graphicspath` and a Pandoc `resource-path` so PDF export can find them even if you run Pandoc from the repo root.

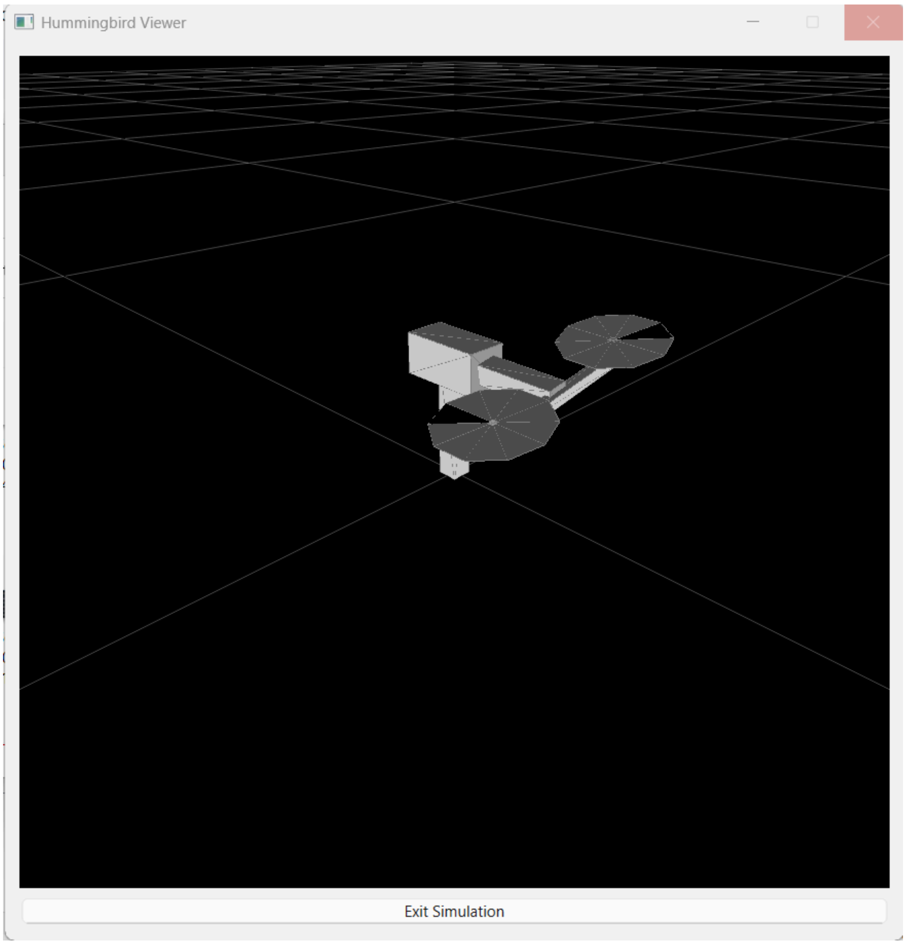
H.7 - Longitudinal PD (Pitch)





H.8 - Lateral PD (Yaw/Roll)





Tip: In matplotlib windows you can use the save icon, or temporarily add `plt.savefig('figs/h7_theta.png', dpi=200, bbox_inches='tight')` near the end of the script.

Repo Map (for this lab)

- Controllers: `ctrlLonPD.py`, `ctrlPD.py`
- Sims/plots: `h7_hummingbirdSim.py`, `h8_hummingbirdSim.py`, `dataPlotter.py`
- Dynamics/params: `hummingbirdDynamics.py`, `hummingbirdParam.py`, `signalGenerator.py`
- This report: `Lab3_Review.md`, lessons: `Lab3_Learned.txt`

Troubleshooting

- If you see OpenGL/pyqtgraph warnings during animation, they are typically harmless on headless or constrained GPUs.
- If `_hummingbird_sim` imports fail when using `-m`, run from inside the folder with `python h7_hummingbirdSim.py`.